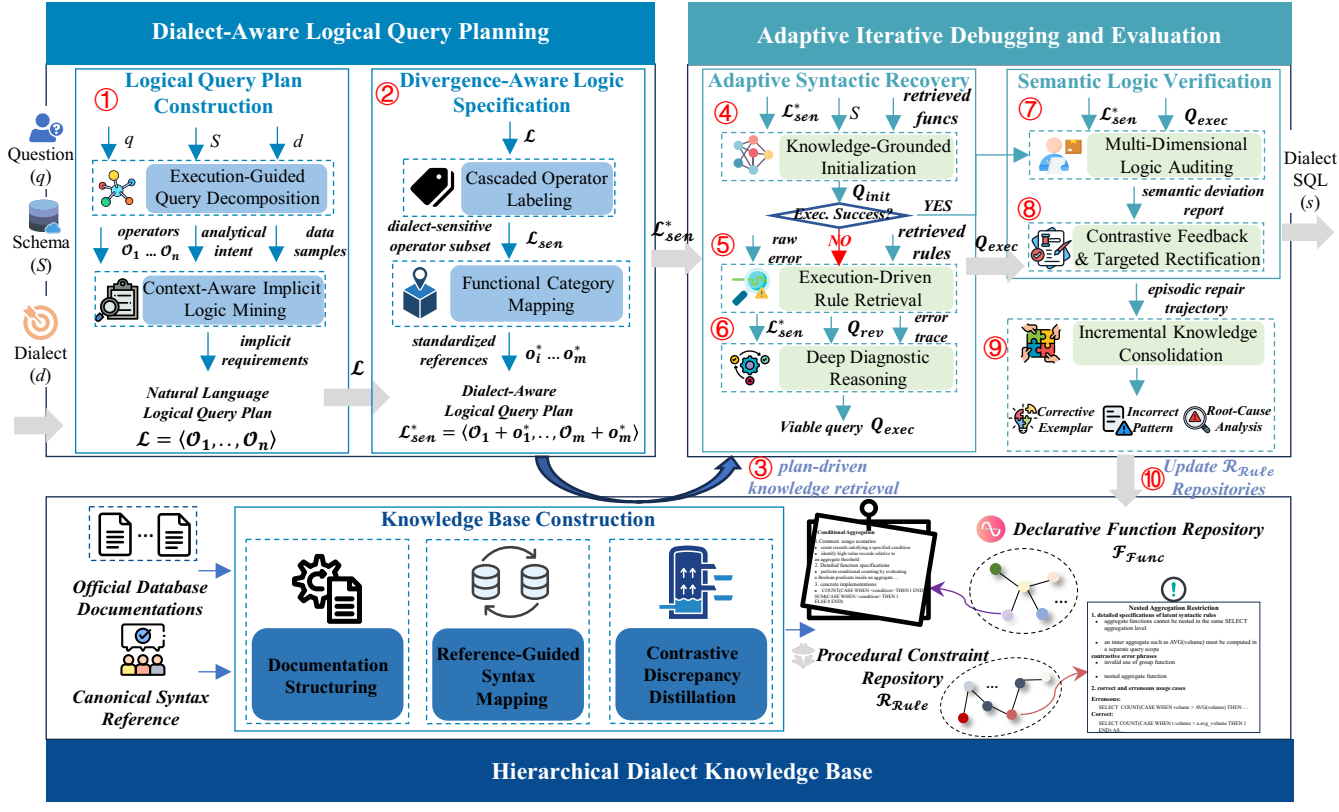

Dialect-Aware Logical Query Planning
Adaptive Iterative Debugging and Evaluation
Question (q)
Schema (S)
Dialect (d)
① Logical Query Plan Construction
q
S
d
Execution-Guided Query Decomposition
operators $O_1 \dots O_n$
analytical intent
data samples
Context-Aware Implicit Logic Mining
implicit requirements
Natural Language Logical Query Plan
$\mathcal{L} = \langle O_1, .., O_n \rangle$
$\mathcal{L}$
② Divergence-Aware Logic Specification
$\mathcal{L}$
Cascaded Operator Labeling
dialect-sensitive operator subset
$\mathcal{L}_{sen}$
Functional Category Mapping
standardized references
$o_i^* \dots o_m^*$
Dialect-Aware Logical Query Plan
$\mathcal{L}^*_{sen} = \langle O_1 + o_1^*, .., O_m + o_m^* \rangle$
$\mathcal{L}^*_{sen}$
Adaptive Syntactic Recovery
④ $\mathcal{L}^*_{sen}$
S
retrieved funcs
Knowledge-Grounded Initialization
Q_{init}
Exec. Success?
YES
NO
⑤ raw error
retrieved rules
Execution-Driven Rule Retrieval
⑥ $\mathcal{L}^*_{sen}$
Q_{rev}
error trace
Deep Diagnostic Reasoning
Viable query Q_{exec}
Q_{exec}
Semantic Logic Verification
⑦ $\mathcal{L}^*_{sen}$
Q_{exec}
Multi-Dimensional Logic Auditing
semantic deviation report
⑧ Contrastive Feedback & Targeted Rectification
episodic repair trajectory
⑨ Incremental Knowledge Consolidation
Corrective Exemplar
Incorrect Pattern
Root-Cause Analysis
Dialect SQL (s)
③ plan-driven knowledge retrieval
⑩ Update $\mathcal{R}_{Rule}$ Repositories
Official Database Documentations
Canonical Syntax Reference
Knowledge Base Construction
Documentation Structuring
Reference-Guided Syntax Mapping
Contrastive Discrepancy Distillation
Conditional Aggregation
Declarative Function Repository $\mathcal{F}_{Func}$
Procedural Constraint Repository $\mathcal{R}_{Rule}$
Nested Aggregation Restriction
Hierarchical Dialect Knowledge Base